Day 44 – Cloud Connector

Class date: 1st September 2026 | Module: Connectivity (SAP BTP Cloud Connector)

Clarification made to your handwritten notes: you labelled the two connection directions as **(1) Inbound request – CPI to ICICI Bank** and **(2) Outbound request – ICICI to CPI, process connection.** Point (1) is correct. Point (2), however, was mixing up two different things: the reply that ICICI Bank's system sends back to CPI is simply the **response** to CPI's own request (it's not a separate "outbound request" in its own right). The genuine **Outbound** scenario – matching the Google.com example you drew right below it – is when a system **inside** ICICI's network wants to reach something **outside** on the public internet (e.g. ICICI's own server calling google.com). That is handled by the Cloud Connector's separate **HTTP(S) Proxy** feature, and it has nothing to do with CPI. This distinction is explained fully in Sections 2 and 4 below.

1. What is a Firewall? (start here first)

A **Firewall** is a security control that operates at the **network layer**. Its job is to make sure data stays secure by only allowing trusted, explicitly-permitted traffic to enter (or leave) a network, and blocking everything else by default.

Example used in class: Infosys runs an **ICICI Bank** system (i.e. Infosys, as an implementation partner, manages banking systems for ICICI). If any **outside system** simply tries to call directly into ICICI Bank's network, the **firewall** sitting in front of that network blocks the request by default – nothing gets in unless it has been explicitly whitelisted.

2. The problem, from SAP CPI's point of view

Picture two on-premise systems: **System 1** (a regular, directly-reachable system) and **System 2** (a system that lives inside ICICI Bank's protected network). SAP CPI, running on the public cloud (BTP), can talk to System 1 without any special setup. But when CPI tries to reach System 2 sitting behind ICICI Bank's firewall, the **firewall blocks the connection** – it does not trust or allow unknown incoming calls from CPI (or from any other external caller) by default.

The two connection directions to keep straight:

- **Inbound request:** CPI → ICICI Bank. This is CPI (cloud) trying to reach a system that sits inside ICICI's on-premise network. From ICICI's network's point of view, this traffic is coming *inward* – hence "inbound." This is the scenario a firewall blocks by default.
- **Outbound request:** a system *inside* ICICI's own network reaching *out* to the public internet (e.g. to google.com) – this is a different scenario altogether, covered in Section 4.

So the core problem to solve is: how does CPI (outside, on the cloud) securely reach System 2 (inside ICICI's firewalled network), **without** ICICI's network team having to open a risky inbound port on their firewall just for this purpose?

3. Cloud Connector – the solution

Cloud Connector is a **software** product from SAP (a free SAP product, Java-based) that is **installed inside the ICICI network itself** – i.e. on ICICI's own infrastructure, behind their own firewall.

How it solves the problem: instead of CPI trying to push a connection *into* ICICI's network (which the firewall blocks), the Cloud Connector inside ICICI's network initiates a **secure, encrypted tunnel**

outward to SAP BTP's Connectivity service. Since this is an **outbound** connection from ICICI's own side, it does not require the firewall to open any new inbound ports – outbound connections are typically already allowed. Once this tunnel exists, CPI can send requests through it to reach System 2, and the traffic travels through the already-open, secure tunnel rather than needing a new direct inbound hole in the firewall. (This approach is officially called **"reverse invoke."**)

Setup requirements, as noted in class: the Cloud Connector needs to know which **System/Server** it should expose, and it must be linked to **SAP BTP subaccount details** (Cloud Connector only supports connecting to **SAP BTP subaccounts** – it is tied to one or more specific subaccounts, not a generic open gateway).

Important practical detail (from official SAP documentation, useful to remember): when configuring the Receiver adapter in CPI to call an on-premise system via Cloud Connector, the **Address field must always use the *HTTP* protocol, never HTTPS** – even if the actual target system on the ICICI side uses HTTPS. This looks unusual at first, but it's correct: the data is still fully encrypted, because it travels through the secure tunnel between BTP and the Cloud Connector; the Cloud Connector itself then manages the final HTTPS hop to the real backend system based on the system mapping configuration.

4. Firewall Proxy Setup – the true "Outbound" scenario

This is a separate concept from Section 3, and is usually handled by ICICI's own **internal network team**, not by the CPI/integration team.

Example: ICICI's own server (inside their network) wants to reach an external, public internet resource – e.g. **google.com**. Since this call is going *out* of ICICI's network to the internet, it is a genuine **outbound** request. A **Proxy** is configured for this purpose, sitting between ICICI's internal systems and the public internet.

Proxy Setup is only used for outbound connections – i.e. traffic initiated from inside ICICI's network going out, such as the ICICI-server-to-google.com example. This proxy is what **temporarily bypasses** the strict firewall rules for that one permitted outward path, without opening the network up more broadly.

5. Secure Tunnel & SAP Sub-Accounts

The connection between the Cloud Connector and SAP BTP is established as a **Secure Tunnel**. This tunnel **only supports SAP Sub-Accounts** – meaning a given Cloud Connector instance is registered against one or more specific BTP subaccounts, and only integration flows running under those registered subaccounts can route traffic through it.

Putting it all together (final flow, as drawn in class):

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CPI --(Secure Tunnel)--> Cloud Connector --(installed inside ICICI's network)--> SAP S/4HANA @ ICICI Bank Response comes back: SAP S/4HANA @ ICICI Bank --(HTTP, outbound through the same tunnel)--> CPI In the CPI Receiver adapter, this is configured using: Proxy Type = On-Premise, plus the Proxy Host and Proxy Port values provided by the BTP Connectivity service (these route the call through the correct Cloud Connector instance).
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Quick-recall summary

- **Firewall** = network-layer security that blocks untrusted traffic by default (both inbound and outbound unless explicitly allowed).

- **Inbound (CPI → ICICI Bank)** = cloud calling into the on-premise network – this is what Cloud Connector solves.
- **Outbound (ICICI server → google.com)** = the on-premise network's own systems calling out to the internet – a separate HTTP(S) Proxy feature, handled by the network team.
- **Cloud Connector** = free SAP software, installed inside the customer's own network, that opens a **secure outbound tunnel** to a specific SAP BTP subaccount ("reverse invoke") – no inbound firewall ports need to be opened.
- In the CPI Receiver adapter: Proxy Type = **On-Premise**; Address protocol must be **HTTP**, never HTTPS, even though the traffic remains encrypted end-to-end via the tunnel.